Tutorial 3 (to be completed in week3)
ER Modelling

Tutorial Objectives:

Create an ERD (using J Developer/Visio) that can be implemented for a medical clinic for the given business rules.

Tutorial Tasks:

A. Create an ERD for a government-funded medical clinic based on the given business rules:

i. A patient can make many appointments with one or more doctors (patient can see a different doctor in their next visit) in the clinic, and a doctor can accept appointments with many patients. However, each appointment is made with only one doctor and one patient.

ii. Emergency cases do not require an appointment. However, for appointment management purposes, an emergency is entered in the appointment book as "unscheduled."

iii. If kept, an appointment yields a visit with the doctor specified in the appointment. The visit yields a diagnosis and, when appropriate, treatment.

iv. With each visit, the patient's records are updated to provide a medical history.

v. Each patient visit creates a bill. Each patient visit is billed by one doctor, and each doctor can bill many patients.

vi. Each bill must be paid. However, a bill may be paid in multiple instalments, and a payment covers only one bill.

Additional (Optional) requirements:

vii. A patient may pay the bill directly, or the bill may be the basis for a claim submitted to an insurance company.

viii. If the bill is paid by an insurance company, the deductible (if any) is submitted to the patient for payment.

Your ERD should include the required entities, relevant attributes (also whether optional or not), etc. The diagram should also include the cardinalities, association names and should flag the PKs, FKs, and derived attributes (if any).

State your assumptions (if necessary) in a concise manner but at the same time do not try to alter / increase the scope based on your knowledge of some medical clinic.

Note: As the focus is on getting the modelling right, suggest ***first hand draw your diagram and understand conceptually*** before you create this using a case tool i.e., J Developer/Visio, etc.

The above case is slightly modified from the original problem 8, Ch4, Pg 157 ed 13.

B. Now use your diagram to identify whether:
   - the entities are existence-dependent or not?
   - the relationship degrees are unary, binary, or ternary?
   - the relationship strengths are weak or strong?
   - the entities are weak or strong?
   - the relationship participation between entities is optional or mandatory?
   - any of the primary keys are surrogate keys or composite keys?

C. Further Discussion: Minimum bill amount per patient visit must be $75. Government regulations also advice that this bill amount cannot exceed $250. Where and how will you model these business requirements?